

Artificial Intelligence Handbook - Part 1

Artificial Intelligence

Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence such as learning, reasoning, planning, and decision making.

Machine Learning

Machine Learning is a subset of AI where models learn patterns from data. Algorithms include Linear Regression, Decision Trees, Random Forests, SVMs, and Neural Networks.

Deep Learning

Deep Learning uses multiple neural network layers. Common architectures include CNNs, RNNs, LSTMs, and Transformers.

Natural Language Processing

NLP allows computers to understand and generate human language. Applications include chatbots, translation, summarization, and sentiment analysis.

Computer Vision

Computer Vision processes images and videos for image classification, object detection, OCR, and face recognition.

Python Libraries

Popular libraries are NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, OpenCV, ChromaDB, LangChain, and Sentence Transformers.

RAG

Retrieval-Augmented Generation retrieves relevant document chunks from a vector database before generating an answer.

Vector Database

ChromaDB stores embeddings and performs semantic similarity search.

Best Practices

Use clean documents, smart chunking, overlap, high-quality embeddings, and validate retrieved context.

Interview Questions

What is AI? Explain RAG. What is ChromaDB? Compare TensorFlow and PyTorch. Explain semantic search.

Artificial Intelligence Handbook - Part 2

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